

MVP SOC Infrastructure Analysis, Advice & Design

VINO AI for Cybersecurity

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Introduction

In the scope of the realization of SOC MVP for VINO AI a network infrastructure was designed. This infrastructure supports and extends the original plan in applied research. By employing this infrastructure practical red and blue team operations and exercises can be conducted and analysed. The purpose is to support research & development of AI guided

SOC. Information in this document must allow users to reproduce the system and conduct operations intended by the main program. The product is targeting Small and medium Enterprises (SME) and aligns with goals of Fontys Cybersecurity Research Group.

Background

In the past several attempts to formulate the problem and plan the project were performed [1]. Current documentation expands previous work.

Functional & Non-Functional Requirements

Category	Requirement
FR 1: Telemetry & Logging	Windows VM must forward Sysmon/Windows Event logs to Wazuh Manager over an encrypted channel.
FR 2: Adversary Emulation	Caldera server on Linux host must send “abilities” with agents (e.g. Manx/Sandcat) to Windows VM to execute operations with MITRE ATT&CK TTPs.
FR 3: Remote Control	Administrator’s Mac laptop must connect to Linux host via Cisco VPN and SSH for Wazuh/Caldera management and server control. Administrators must be able to control the target VM and manage the VSphere environment, both virtual machines and their networking.
FR 4: Firewall Enforcement	Host VM must enforce firewall rules organizing which TCP/UDP ports are open to which source/destination.
FR 5: SOC Prototype Integration	Ability to forward Wazuh alerts (.log, JSON output) into an ELK stack or equivalent for dashboarding.
NFR 1: Secure Communication	All log and command traffic (Wazuh, Caldera, SSH) must be encrypted: VPN for network transport; SSH for host access; HTTPS for Wazuh dashboard.
NFR 2: Modularity & Reproducibility	Infrastructure components (VMs, agents, rulesets) must be documented so the environment can be rebuilt consistently.

NFR 3: Performance & Availability	Host VM (Wazuh, Caldera) should be responsive to on-demand emulation tasks, with CPU/RAM sized appropriately; logging must not drop events under typical workload.
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Infrastructure Analysis

Topological Mapping Review

Node	Role	Location / IP	Notes
MacBook (Local)	Admin + dashboard	Dynamic IP via VPN (192.168.223.0/24)	Runs SSH client, receives tunneled Caldera/Wazuh UI
VPN Gateway	Cisco VPN concentrator	Dynamic IP pool → MacBook	Tunnel interface used to restrict access to Host only
Linux Host VM	Core node: Wazuh, Caldera, Snort	192.168.142.128	SSH entry point; Caldera/Wazuh servers + sniffer port (ens192)
Windows VM	Target node (agent only)	192.168.142.5	Wazuh agent, Caldera agent; isolated except from Host

Firewall & Access Zones

Traffic	Protocol/ Port	Source	Destination	Allowed?	Comment
SSH (admin access)	TCP 22	192.168.223.0/24	Linux Host	✓	Controlled via UFW
Wazuh agent to manager	TCP/UDP 1514/1515	Windows VM	Linux Host	✓	TLS-secured telemetry
Caldera agent to server	TCP/UDP 8888	Windows VM	Linux Host	✓	Reverse HTTPS or local listener
Admin browser (Kibana/Wazuh GUI)	Tunnel → 5601 (via VPN)	MacBook → SSH tunnel	localhost on Linux Host	✓	Via port forwarding

Snort monitoring (ens192)	Passive	--	--	✓	No stack interaction
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UFW Firewall rules

To	Action	From
1514, 1515/tcp	ALLOW	Anywhere
1514/udp	ALLOW	Anywhere
1514/udp	ALLOW	192.168.142.5
1515/tcp	ALLOW	192.168.142.5
22	ALLOW	192.168.223.0/24
8888/tcp	ALLOW	192.168.142.5
8888/udp	ALLOW	192.168.142.5
1514,1515/tcp (v6)	ALLOW	Anywhere (v6)
1514/udp (v6)	ALLOW	Anywhere (v6)

Data-Flow Analysis

Method: Logical Data-Flow Mapping

Goal: Show how data moves through the system

Log Flow – Blue Team (Wazuh)

1. **Event Generation:** Sysmon on Windows logs events (e.g., process creation, file changes).
2. **Agent Forwarding:** Wazuh agent sends logs via TLS over TCP 1514 to Linux Host.
3. **Wazuh Manager Processing:** On Host, logs are parsed, correlated, and alerts are generated.
4. **Optional Visualization:** Alerts accessed via Wazuh Dashboard UI.

Attack Control Flow – Red Team (Caldera)

1. **Operator Access:** Admin tunnels Caldera GUI to MacBook (port 8888 via SSH).
2. **Ability Deployment:** Caldera server sends “ability” to agent on Windows via HTTPS 8888.
3. **Execution:** Windows agent performs action (e.g., list users), potentially triggering logs.
4. **Detection Loop:** Resulting activity is logged → Wazuh detects and forwards alert.

Monitoring Channel

- **Snort IDS:** Linux Host listens on passive interface `ens192`, capturing traffic on the isolated network for detection/logging.

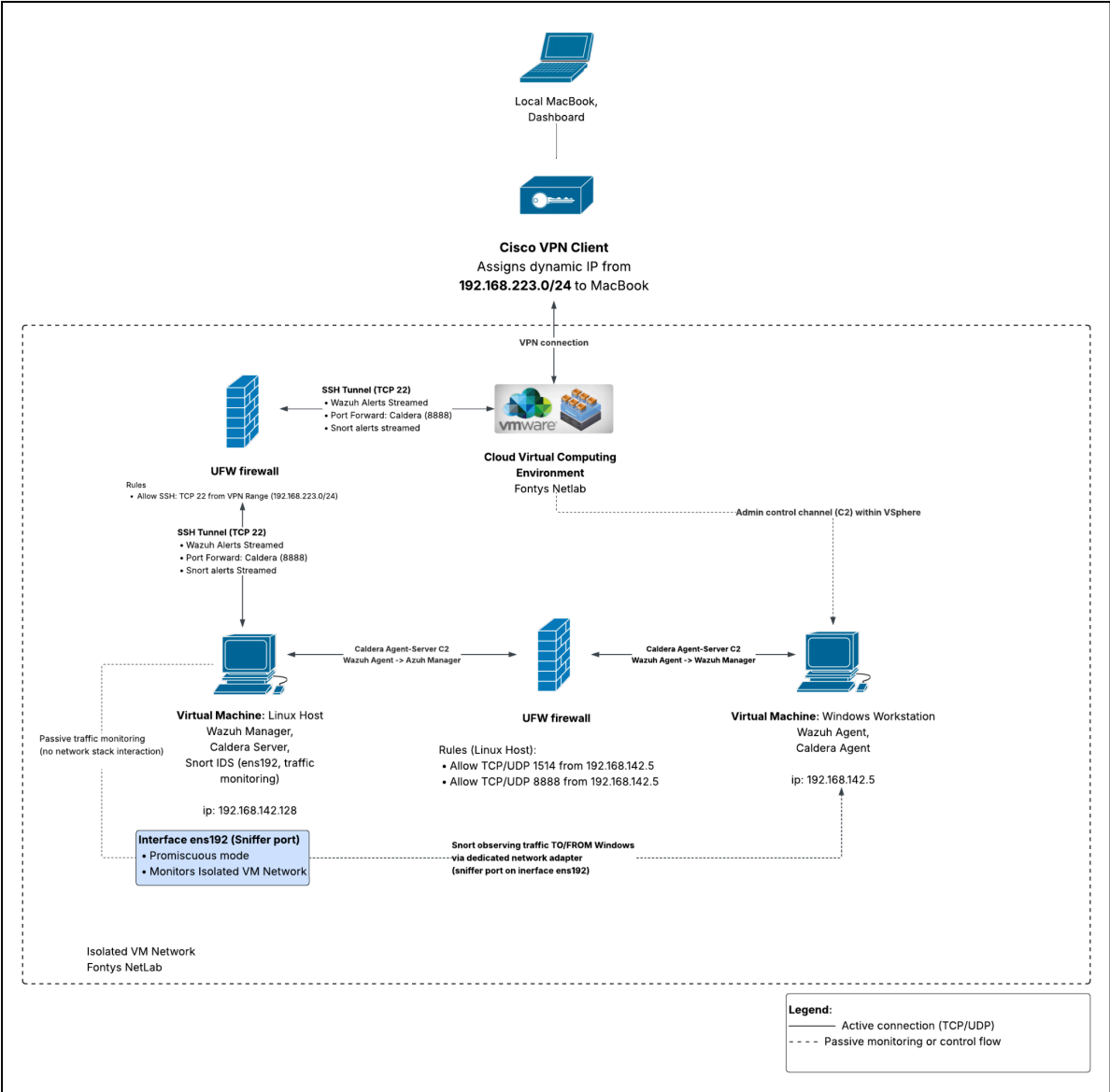
Layered Architecture Model

Method: 3-Tier Logical Architecture

Goal: Classify infrastructure components by function and interaction

Layer	Components	Role
Presentation Layer	MacBook (Admin GUI), SSH tunnel, custom Streamlit dashboard browser	Access layer for operator interaction
Logic Layer	Linux Host: Wazuh Manager, Caldera Server, SSH service	Core logic processing: detection, alerting, simulation
Data Layer	Windows VM: Sysmon logs, Caldera agent, Wazuh agent, network traffic	Raw telemetry sources (events), passive network log capture

Infrastructure Design



MVP SOC Logical & Security Data Flow Diagram

Component specification

Linux Host

Virtual Machine Details

ACTIONS ▾

Power Status

 Powered On

Guest OS

 Ubuntu Linux (64-bit)

VMware Tools

Running, version:12389 (Guest Managed) 

DNS Name (1)

student-vm-ubuntu22

IP Addresses (4)

192.168.142.128
172.18.0.1
[AND 2 MORE](#)

Encryption

Not encrypted



VM Hardware

CPU

4 CPU(s), 59 MHz used

Memory

8 GB, 0 GB memory active

Hard disk 1

64 GB | Thin Provision 
[NIM01-6](#)

Network adapter 1 (of 2)

[0142_Internet-DHCP-192.168.142.0_24](#) (connected) | 00:50:56:97:19:f6

CD/DVD drive 1

Disconnected  ▾

Compatibility

ESXi 8.0 U2 and later (VM version 21)

Windows target

Virtual Machine Details

ACTIONS ▾



Power Status	 Powered On
Guest OS	 Microsoft Windows 10 (64-bit)
VMware Tools	Running, version:12421 (Current) ⓘ
DNS Name (1)	DESKTOP-LB9TSLA
IP Addresses (2)	192.168.142.5 fe80::4364:7c24:7376:8c43
Encryption	Not encrypted



VM Hardware

CPU	2 CPU(s), 59 MHz used
Memory	8 GB, 1 GB memory active
Hard disk 1	90 GB Thin Provision ⓘ NIM01-9
Network adapter 1 (of 2)	0142_Internet-DHCP-192.168.142.0_24 (connected) 00:50:56:97:29:61
CD/DVD drive 1	Disconnected  ▾
Compatibility	ESXi 7.0 U2 and later (VM version 19)

EDIT

Advice & Recommendations

Area	Recommendation	Why? / Justification
1. VPN Access Security	Implement two-factor authentication (2FA) for Cisco VPN, e.g., with Duo or FortiToken.	Current setup only uses cert/IP auth; 2FA reduces risk of stolen VPN credentials.
2. SSH Hardening	Enforce key-based login only and disable password login in <code>sshd_config</code> .	Ensures only registered developers can connect. Prevents brute-force.
3. Certificate Rotation	Automate renewal of TLS certificates used by Wazuh (manager/agent) every 90 days.	Prevent cert expiration and support key hygiene.
4. Caldera Isolation	Consider running Caldera in a separate container or VM , not alongside Wazuh.	Reduces attack surface and keeps red-team tooling isolated.
5. Snort Configuration	Load custom Snort rules for detecting C2 patterns and log exfiltration behavior.	Snort is passive, but can add visibility for advanced traffic signatures (even in R&D).
6. Wazuh Rule Expansion	Extend detection coverage by adding rules for fileless execution, LOLBins, PowerShell abuse , etc.	Existing rules detect basic process actions. Add for stealthy TTPs.
7. Logging Segregation	Use separate log directories (or filebeats with tags) for Red Team vs Blue Team logs.	Helps clarify telemetry origin and avoid false positives during testing.
8. Backup Strategy	Add regular config and log backup for Wazuh, firewall rules, and Snort rules (daily or weekly via cron + rsync).	Enables fast recovery and reproducibility.
9. Documentation	Create a versioned, public-facing Git repo with all UFW, agent configs, install scripts, and a setup guide (README).	Aligns with NFR: reproducibility. Also useful for future auditors or teammates.
10. Encryption	Encrypt virtual machines.	Protects Against Disk Theft or Snapshots. If someone gains access to the virtual disk file (.vmdk, .qcow2, .vhdx, etc.), they cannot read its contents without the decryption key.

Management & Control

Standard operating procedures (SOPs), configuration records, and repeatable steps.

System Setup & Configuration Procedures

Component	Procedure
Linux Host VM	- Install Ubuntu (22.04) - Set static IP: 192.168.142.128 - Install Wazuh, Caldera, Snort - Configure UFW with listed ports
Windows Target VM	- Install Windows 10 Pro / Server 2019 - Install Wazuh agent and deploy Caldera agent (e.g., Sandcat) - Set static IP: 192.168.142.5 - Install Sysmon and configure it and agent to send logs
Admin Laptop	- Install Cisco AnyConnect - Set up SSH keys - Configure SSH tunnels to Wazuh and Caldera

Network & Firewall Configuration

Use ufw on linux. Enable, set rules and check.

#Install UFW

```
sudo apt install ufw
```

Enable UFW if not already

```
sudo ufw enable
```

Allow SSH only from your Mac IP (use ip in cisco vpn)

```
sudo ufw allow from <CISCO_VPN_IP (e.g. x.x.x.0/24)> to any port 22 proto tcp
```

Allow Wazuh Agent communication (usually port 1514/udp, 1515/tcp)

```
sudo ufw allow from <WAZUH_AGENT_IP> to any port 1514 proto udp
```

```
sudo ufw allow from <WAZUH_AGENT_IP> to any port 1515 proto tcp
```

#Allow Caldera ssh tunnel and agent communication

```
sudo ufw allow from 192.168.142.5 to any port 8888 proto tcp
```

Deny everything else incoming by default

```
sudo ufw default deny incoming
```

```
sudo ufw default allow outgoing
```

Check status

```
sudo ufw status verbose
```

Routine Management Tasks

Task	Schedule	Responsible Role	Tool
Restart Wazuh service	On alert drop	Admin	systemctl
Update Caldera abilities	Weekly	Red Team operator	Git pull
Backup Wazuh config & rules	Daily	Admin	cron + rsync
Log rotation / archival	Weekly	Admin	logrotate
Snort rule updates	Monthly	Network Security Lead	Pulled from Snort community

Access & Authentication Controls

- VPN access protected with certificate + 2FA (planned)
- SSH access restricted to key-based auth (planned)
- Users: admin, red_team, soc_user
- All admin actions logged using .bash_history and/or auditd (recommended addition)

Documentation & Version Control

- All setup scripts, firewall configs, detection rules are tracked in Git repo
- README contains rebuild instructions for Host and Target from clean base images

Github: https://github.com/Krasnomakov/soc_config

Conclusion

The documented infrastructure represents a complete and modular prototype for a research-oriented SOC environment. By integrating open-source telemetry and adversary emulation tools within a secure and isolated virtual environment, it enables controlled experimentation, monitoring, and alerting workflows aligned with cybersecurity operational needs. Each system component has been configured with a clear role, secured through firewall rules and access controls, and documented to ensure reproducibility and maintainability.

The use of VPN tunneling, encrypted communications, and strict role separation ensures confidentiality and integrity of telemetry data, while the layered architecture promotes scalability. The combination of automated detection (via Wazuh), emulated threat generation (via Caldera), and passive traffic inspection (via Snort) provides a strong foundation for iterative SOC capability development. Management tasks and configuration standards further enable long-term use and controlled evolution of the environment.

This infrastructure serves both as a functional research platform and a learning environment for cybersecurity tooling, analysis, and systems thinking. It is well-positioned for future enhancements, including real-time dashboards, detection engineering, and integration with AI-based reasoning components.

Reference

1. Kraskov A. (2025). VINO AI for Cybersecurity, Project Proposal
Canvas:
https://fhict.instructure.com/courses/13084/assignments/260587?module_item_id=1313006
PDF:  VINO for Cybersecurity Project Proposal v2.pdf